



DEALER BEST PRACTICE SERIES

Kits for Dynamometer Test

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

The Product Support Division of Ferreyros is continuously searching for ways to optimize the cycle time of component rebuilds in the CRC using CPS processes. Improving component turnaround time increases efficiencies, capacity, and lowers cost resulting in higher levels of customer satisfaction.

In the dyno test process, there was an opportunity to improve the set-up times, ordering process, and developing standardized set up kits, thus avoid unnecessary time lost looking for these materials.

2.0 Best Practice Description

CRC Dynamometer had their hoses and accessories storage in one place but without identification. That process generates excessive search times to find connecting tooling required. According to the measurement performed in December, the time to look for and hook up accessories was out of standards.

To remove the waste mentioned in this area, accessory parts kits were created for each model of engine in order to install the engine to the dyno, color-coded for identification, and stored organized by engine model.

3.0 Implementation Steps

- I. The Kickoff Meeting and formation team.



Figure 1: Team work

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II. Determine frequency of engine models tested in dynamometers (Pareto).

- Historical data was used to determine the most frequent engine models.
- Then, an engine classification was performed to identify subclasses of engine.
- Finally, a Pareto analysis was made to focus on the 80% most common engines.

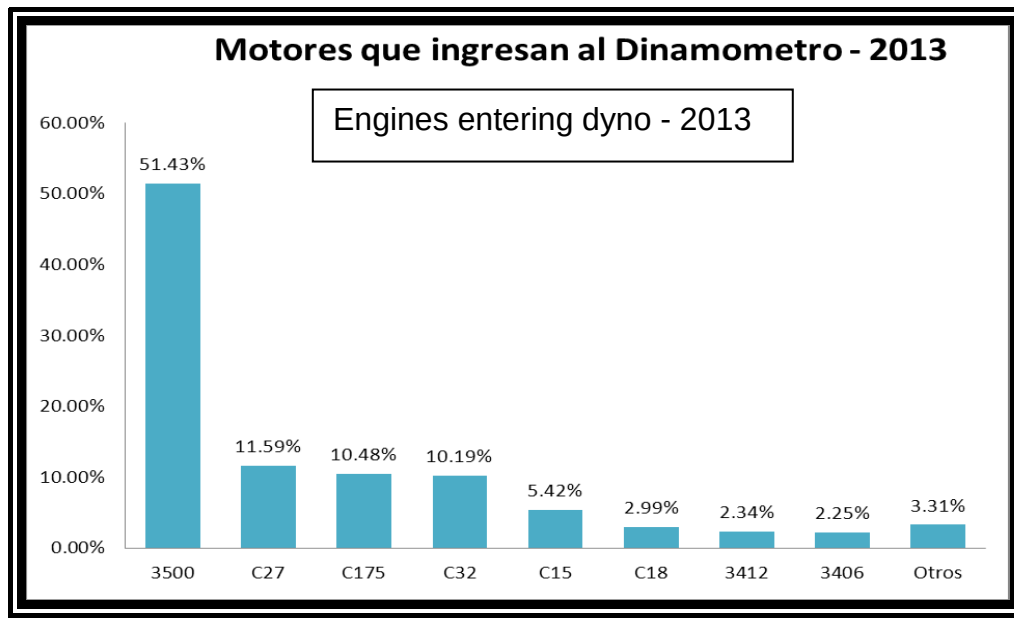


Figure 2 – Pareto of engines.

III. Quality circle to determine the list of kits to develop.

Quality circle was made with the team to determine the list of kits required: hoses, exhaust pipes, thermocouples and connectors.

a. Water hoses:

Sala means “Dyno Room”

NOMBRE	MODELO	SALA
KIT 1	3500 (793C - 793D)	SALA 3
KIT 2	C175 (793F - 797F)	SALA 3
KIT 3	3500 (785C)	SALA 1 - SALA 2
KIT 4	C18 - C15	SALA 2
KIT 5	3500 (793C - 793D)	SALA 1
KIT 6	C27 - C32	SALA 2
KIT 7	UNIVERSAL	VARIOS
KIT 8	Motores Marinos	VARIOS

Figure 3 –Kits of water hoses.

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b. Exhaust Pipes:

NOMBRE	DESCRIPCIÓN
KIT 1	Para motores 3500
KIT 2	Para motor C175 High
KIT 3	Para motor C175 Low
KIT 4	Para motores universal

Figure 4 –Kits of exhaust pipes.

c. Hoses::

Nombre	Descripción
KIT 1	SALA 1
KIT 2	SALA 2
KIT 3	SALA 3

Figure 5 –Kits of dialysate and Fuel hoses

d. Thermocouples and connectors (make sure thermocouples are properly calibrated):

NOMBRE	MODELO	SALA
KIT 1	3500	SALA 1
KIT 2	C32 - C27 - C18 - C15	SALA 1
KIT 3	C32 - C27 - C18 - C15	SALA 2
KIT 4	3306	SALA 2
KIT 5	3412 - 3176	SALA 2
KIT 6	C175	SALA 3
KIT 7	3500	SALA 3

Figure 6 –Kits of thermocouples and connectors

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- IV. Implementation:
- Creating kits.



Figure 8 –Hose Kit N°3



Figure 9 –Exhaust Pipes Kit 3500

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Figure 10 – Thermocouple Kit 3412-3176



Figure 11 – Kit of dynamometer 3

- Identification. Parts color coded.



Figure 12 – Identification of hoses

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- Fabricated kit baskets.
Baskets that were designed with the help of technicians for proper storage of hoses and accessories for each type of engine were manufactured.

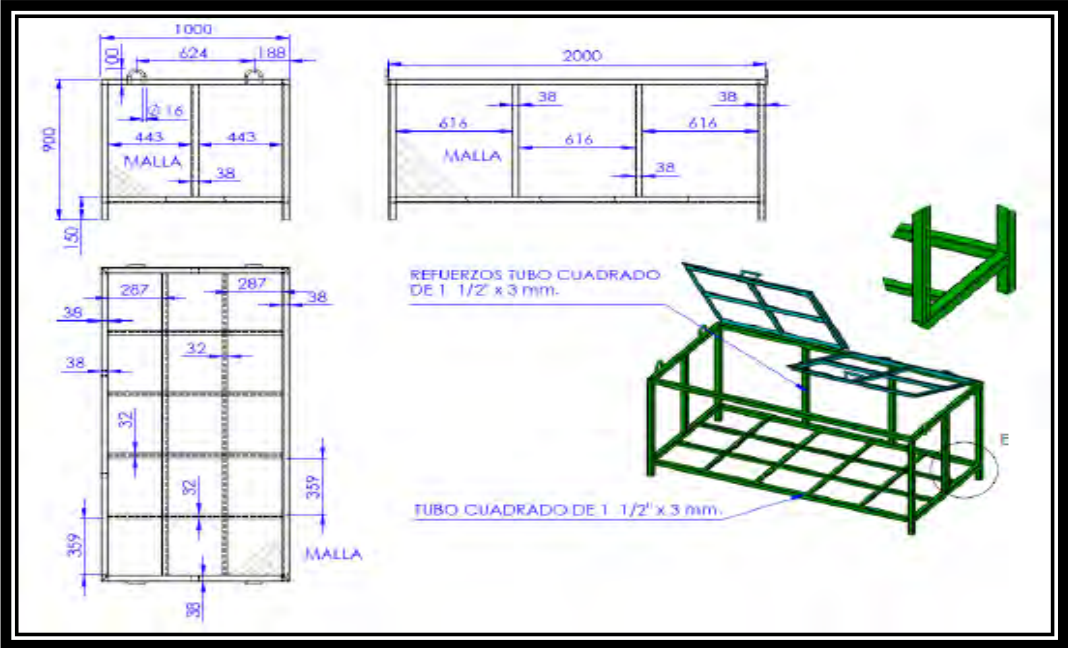


Figure 13 – Baskets were designed by technicians

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Figure 14 – Identification and location of Kits in warehouse

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- Content list / Inventory of each kits.

KIT 1	Descripción	Ubicación	
	Mangueras de Agua - Motores 3500 /793C -	Sala 3	
Código	Descripción	Dimensiones	Imagen
EAM-01	Manguera entrada motor	L: 37"	
		D: 4"	
EAM-02	Manguera entrada motor	L1: 10" L2: 11,5"	
		D1: 5,5" D2: 4"	
EAM-03	Codo 45°		

Figure 15 – Kits inventory

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V. Closing.

Employees were recognized by the company in the journal "Image".

In an exclusive page is described the great work done and the benefits generated for the company.



Figure 16 – Journal "Image"

4.0 Benefits

- ◆ Excess and duplication of accessories and hoses were removed.
- ◆ Search time and installing component for testing was reduced.
- ◆ Excessive time to look for and installation of accessories (about 20% of the complete test) were removed.
- ◆ Work areas are neat and clean
- ◆ Appropriate storage of hoses and accessories, keeping them and avoid being contaminated.
- ◆ Using ergonomics principles, ways of working were improved
- ◆ Reduced test time by WO.

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5.0 Resources Required

- Project sponsor
- Project owner
- CPS team
- Dynamometer technician
- Dealer involvement for contamination control, special tooling and procedures advice.
- Shelving, racking, labeling, paint etc...

6.0 Supporting Attachments / References



Baskets protected with stretch film

Figure 17 – Protection of baskets

EDITOR'S NOTE: Make sure the stretch film used is highly durable to resist tears and allow debris into the container. Installing caps on the hoses is a good idea for extra protection.

7.0 Related Best Practices

N/A

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8.0 Acknowledgements

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